

PCA

1.

Imagine you have a lot of data
— 10 measurements for each student in your class (height, weight, marks in 8 subjects).

Goal of PCA

↳ Reduce the no. of variables (dimensions)

↳ keep as much of the original information (variance) as possible.

→ One can think of PCA as a "data compression" technique" — but without losing much information.

→ Given a set of "data points",
 2. "understand", something useful about them.

Data points → vectors in $\mathbb{R}^d$

$$\begin{bmatrix} \text{height} \\ \text{weight} \\ \text{Age} \end{bmatrix} \in \mathbb{R}^3$$

Problem:-

Input: $\{x_1, x_2, \dots, x_n\}$

$x_i \in \mathbb{R}^d$ d → # of features.

Output! Same "compressed" representation of the dataset.

Example:-

$$\left\{ \begin{matrix} x_1 \\ \begin{bmatrix} -7 \\ -14 \end{bmatrix} \end{matrix}, \begin{matrix} x_2 \\ \begin{bmatrix} 2.5 \\ 5 \end{bmatrix} \end{matrix}, \begin{matrix} x_3 \\ \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \end{matrix}, \begin{matrix} x_4 \\ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{matrix} \right\}$$

$\nearrow \in \mathbb{R}^2$

~~$2 \times n$~~

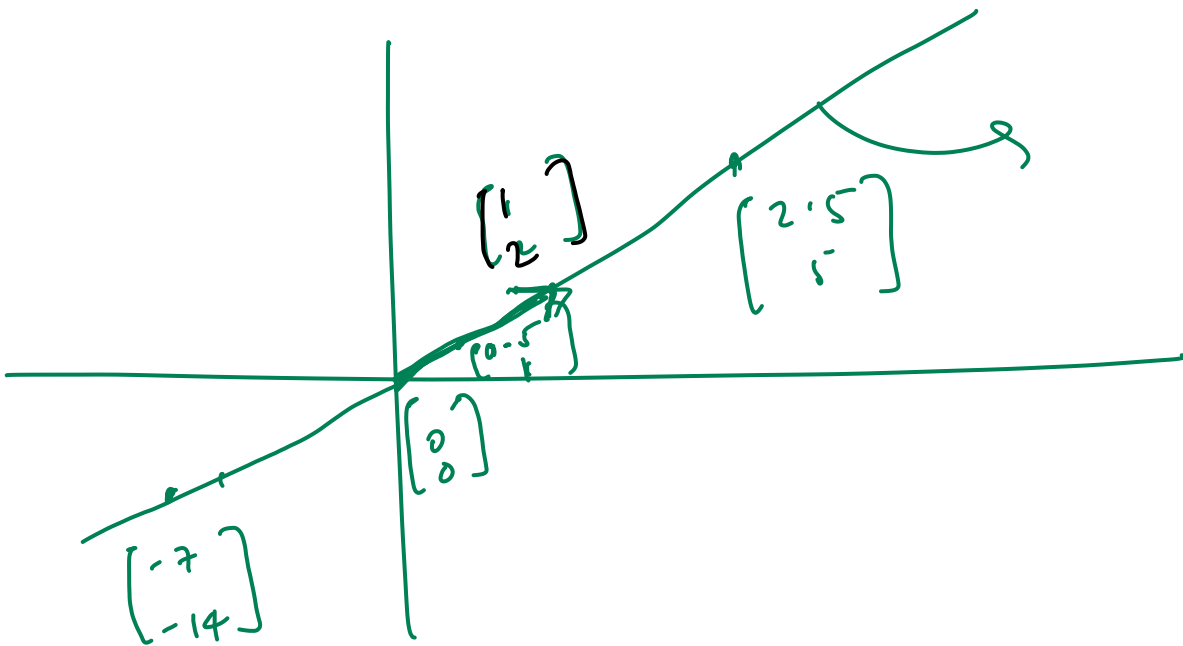
Q:- How many real numbers are needed to store this dataset.

8

Representative	Coefficients
3. $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$	$\{-7, 2.5, 0.5, 0\}$

⑥

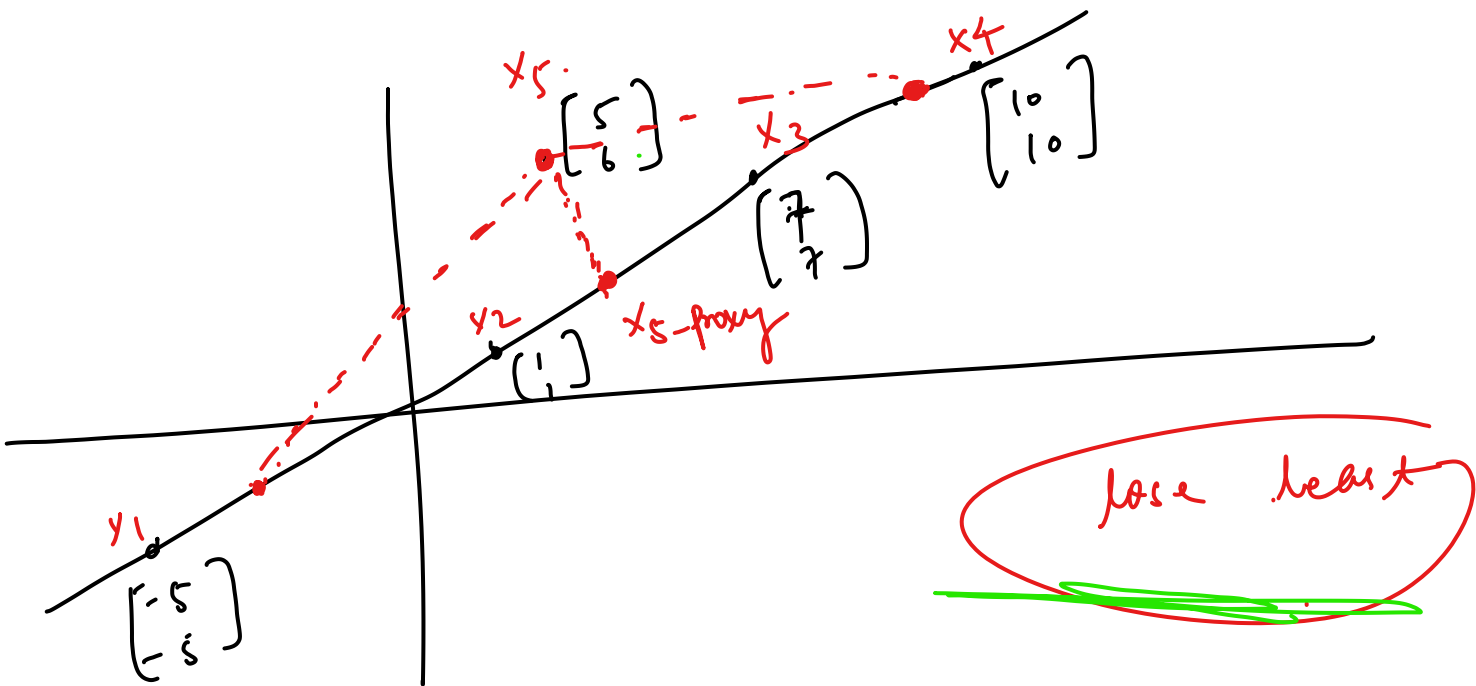
Note: - using representative & coefficients
 = can "reconstruct" the dataset exactly.



Rep.	coeff
$\checkmark \begin{bmatrix} 1 \\ 2 \end{bmatrix}$	$\{-7, 2.5, 0.5, 0\}$
$\checkmark \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$	$\{-7\sqrt{5}, 2.5\sqrt{5}, 0.5\sqrt{5}, 0\}$

original : # real numbers = dn

4. # real numbers in
compressed version = $d + n$

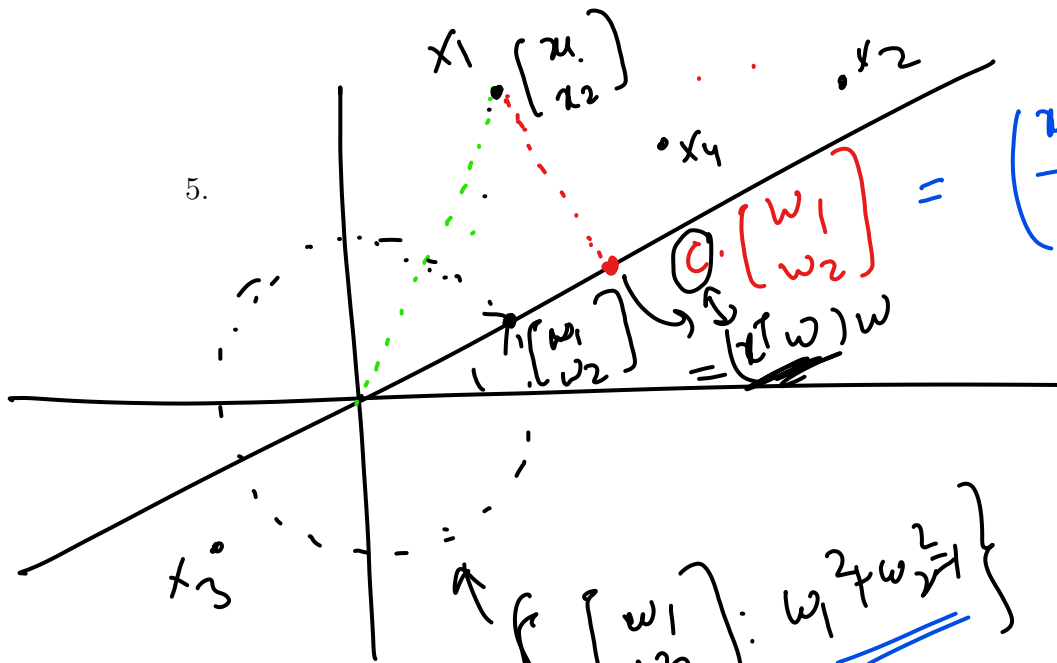


$$\begin{array}{c} \text{Rep} \\ \hline \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \end{array}$$

$$\begin{array}{c} \text{Coef} \\ \hline \{(-5, -5), (1, 1), (7, 7), (10, 10)\} \end{array}$$

$$x_3 = 7 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 7 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix}$$

5.



$$\left(\frac{x_1 w_1 + x_2 w_2}{w_1^2 + w_2^2} \right) \cdot \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

$$= \frac{x^T w}{\|w\|^2} \cdot \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

$$= x^T w \cdot \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

$$= (x^T w) w$$

$$\left\{ \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} : w_1^2 + w_2^2 = 1 \right\}$$

min_c length² (error Vector)

↳

$$\begin{bmatrix} x_1 - c w_1 \\ x_2 - c w_2 \end{bmatrix}$$

$$\min_c (x_1 - c w_1)^2 + (x_2 - c w_2)^2$$

$$c^* = \frac{x_1 w_1 + x_2 w_2}{w_1^2 + w_2^2}$$



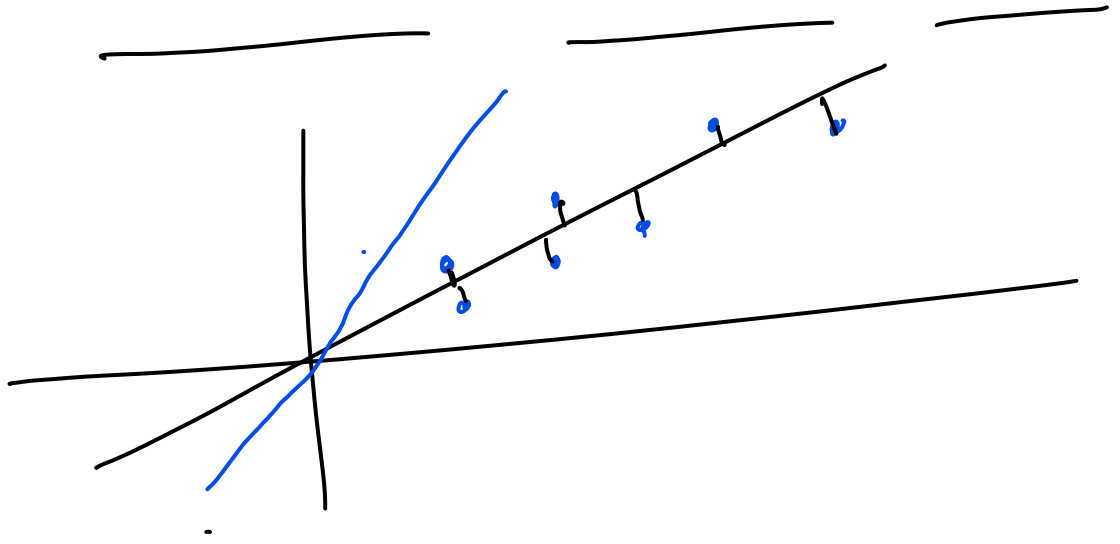
numerator part: dot product of x & w .

denominator part: length² $\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$

Choose $\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ such that $\text{length}\left(\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}\right) = 1$.

6.

Goal: -



→ Find the line that has the least "Reconstruction" error.

Data set: - $\{x_1, x_2, \dots, x_n\}$; $x_i \in \mathbb{R}^d$

$$\text{Error} = \sum_{i=1}^n \text{error}(\text{line}, x_i)$$

$$= \sum_{i=1}^n \text{length}^2 \left(x_i - (x_i^T w) w \right)$$

$$= \sum_{i=1}^n \| x_i - (x_i^T w) w \|^2$$

$$f(w) = \frac{1}{n} \sum_{i=1}^n \|x_i - (x_i^T w) w\|^2$$

$$\|z\|^2 = z^T z$$

$$= \frac{1}{n} \sum_{i=1}^n \left(x_i - (x_i^T w) w \right)^T \left(x_i - (x_i^T w) w \right)$$

solve this

$$= \frac{1}{n} \sum_{i=1}^n \left(\underbrace{x_i^T x_i}_{\text{scalar}} - (x_i^T w)^2 \right) \checkmark$$

$$\min_{\substack{w \\ \|w\|^2=1}} g(w) = \frac{1}{n} \sum_{i=1}^n - (x_i^T w)^2 \checkmark$$

$$\max_{\substack{w \\ \|w\|^2=1}} \frac{1}{n} \sum_{i=1}^n (x_i^T w)^2$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i^T w)^T (x_i^T w)$$

$$= \frac{1}{n} \sum_{i=1}^n (w^T x_i) (x_i^T w)$$

$$= \frac{1}{n} \sum_{i=1}^n w^T (x_i x_i^T) w$$

8.

$$= w^T \underbrace{\left(\frac{1}{n} \sum_{i=1}^n x_i x_i^T \right)}_{C \text{ } d \times d \text{ matrix}} w$$

$$= \max_{\substack{w \\ \|w\|^2=1}} w^T C w$$

$$C = \frac{1}{n} \sum_{i=1}^n x_i x_i^T$$

Covariance matrix

Soln: $w^* = w_1$ is the eigenvector corresponding to the maximum eigenvalue of C .

① $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_k$

Recall:

$$\text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$C = \frac{1}{n} (X - \mu) (X - \mu)^T$$

if the means is not zero.

$$C = \frac{1}{n} X X^T$$

$\alpha_1 = x_i^T w_1 \rightarrow$ Projection of x_i on w_1

Question:- Data points:

9. $x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $x_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $x_3 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$, $x_4 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$

For the given data points, compute the following:

- (i) The mean vector of the data points.
- (ii) The covariance matrix, $C = \frac{1}{n} \sum_i (x_i - \bar{x})(x_i - \bar{x})^T$
- (iii) Eigen values of C ✓
- (iv) Eigen vectors of C ✓
- (v) Transformed data points for one dimensional PCA ✓
- (vi) The reconstruction error for one dimensional PCA. ✓

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad x_3 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad x_4 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

10.

① Mean vector of the data points:-

$$\bar{x} = \frac{1}{4} \begin{bmatrix} 0+2+2+4 \\ 0+2+4+2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

② Covariance matrix, $C = \frac{1}{n} \sum_i \underbrace{(x_i - \bar{x})}_{x_i'} \underbrace{(x_i - \bar{x})^T}_{x_i'^T}$

$$C = \frac{1}{n} \sum_i x_i' x_i'^T$$

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad x_3 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad x_4 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$x_1' = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

$$x_2' = x_2 - \bar{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_3' = \begin{bmatrix} 2 \\ 4 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$x_4' = \begin{bmatrix} 4 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$x_1' x_1'^T = \begin{bmatrix} -2 \\ -2 \end{bmatrix} \begin{bmatrix} -2 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$$

$$x_2' x_2'^T = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$x_3' x_3'^T = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \begin{bmatrix} 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$$

$$x_4' x_4'^T = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix}$$

$$C = \frac{1}{4} \sum_{i=1}^n x_i x_i^T = \frac{1}{4} \{ x_1 x_1^T + x_2 x_2^T + x_3 x_3^T + x_4 x_4^T \}$$

11.

$$= \frac{1}{4} \left\{ \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix} \right\}$$

$$= \frac{1}{4} \begin{bmatrix} 8 & 4 \\ 4 & 8 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Annotations: Arrows point from the diagonal elements 2 and 2 to the word "Var" (circled in green). A large green oval encircles the entire matrix result.

(ii) Eigen values of C.

$$\det(C - \lambda I) = 0$$

$$\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)^2 - 1 = 0$$

$$\Rightarrow 4 + \lambda^2 - 4\lambda - 1 = 0$$

$$\Rightarrow \lambda^2 - 4\lambda + 3 = 0$$

$$\Rightarrow \lambda^2 - 3\lambda - \lambda + 3 = 0$$

$$\Rightarrow \lambda(\lambda - 3) - 1(\lambda - 3) = 0$$

$$\Rightarrow (\lambda - 3)(\lambda - 1) = 0$$

eigenvalues: $\lambda_1 = 3$ $\lambda_2 = 1$

12.

(v) Eigen vectors of C:-

for $\lambda_1 = 3$: solve : $(C - \lambda I)v = 0$
 $\Rightarrow (C - 3I)v = 0$

$$C = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} v = 0$$

$$R_2 \rightarrow R_2 + R_1 \quad \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow -v_1 + v_2 = 0$$
$$\Rightarrow v_1 = v_2$$

eigenvector corresponding to $\lambda_1 = 3$: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$w_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

~~for~~ eigenvector corresponding to $\lambda_2 = 1$

$$(C - I)v = 0$$

$$C = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

13.

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow R_2 \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow v_1 + v_2 = 0$$

$$\Rightarrow v_1 = -v_2$$

eigenvector:

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$w_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Next step:-

Projecting data points

Projecting x_1^1 on w_1 .

→ Projection of x_1^1 on w_1 :

$$\alpha_1 = x_1^{1T} w_1 =$$

$$\begin{bmatrix} -2 & -2 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -2 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\alpha_1 = -2\sqrt{2}$$

14.

Projection of x_2^1 on w_1

$$\alpha_2 = x_2^{1T} w_1 = \frac{1}{\sqrt{2}} [0 \ 0] \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0$$

Projection of x_3^1 on w_1

$$\alpha_3 = x_3^{1T} w_1 = \frac{1}{\sqrt{2}} [0 \ 2] \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \sqrt{2}$$

Projection of x_4^1 on w_1

$$\alpha_4 = x_4^{1T} w_1 = \frac{1}{\sqrt{2}} [2 \ 0] \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \sqrt{2}$$

Transformed data points are calculated as:

$$\tilde{x}_i = (x_i^{1T} w_1) w_1 + \underbrace{(\bar{x}^T w_2) w_2}_{\text{2nd term will vanish.}}$$

$$\begin{aligned} \tilde{x}_1 &= (x_1^{1T} w_1) \cdot w_1 + \bar{x} \\ &= (-2\sqrt{2}) \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} -2 \\ -2 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

during session,
this equation was
incorrect.

Please make a note
of it.

$$\tilde{x}_2 = (x_2^T w_1) w_1 + \bar{x}$$

$$15. = 0 \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \checkmark$$

$$\tilde{x}_3 = (x_3^T w_1) w_1 + \bar{x} = (\sqrt{2}) \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \checkmark$$

$$\tilde{x}_4 = (x_4^T w_1) w_1 + \bar{x} = (\sqrt{2}) \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \checkmark$$

(VI) Reconstruction error:

$$J = \frac{1}{n} \sum_{i=1}^n \|x_i - \tilde{x}_i\|^2$$

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \tilde{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \|x_1 - \tilde{x}_1\|^2 = 0$$

$$x_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \tilde{x}_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \Rightarrow \|x_2 - \tilde{x}_2\|^2 = 0$$

$$x_3 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \tilde{x}_3 = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \Rightarrow \|x_3 - \tilde{x}_3\|^2 = (-1)^2 + 1^2 = 2$$

$$x_4 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \tilde{x}_4 = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \Rightarrow \|x_4 - \tilde{x}_4\|^2 = 1^2 + (-1)^2 = 2$$

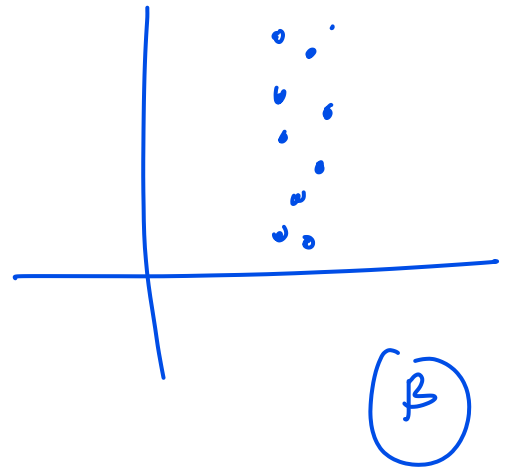
15

$$J = \frac{1}{4} \{ 0 + 0 + 2 + 2 \} = \frac{4}{4} = 1 \checkmark$$

$$\boxed{\text{Projected Variance} = d_1 = 3}$$

16.

↳ The direction which minimizes the "reconstruction error" is exactly the same direction which maximizes the variance



Q.2:- Data points:

17. $x_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, x_2 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$

For the given data points, calculate the followings:-

→ (1) The mean vector of the data points.

→ (2) The covariance matrix, $C = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^T$

→ (3) Eigen values of C

→ (4) Eigen vectors of C

→ (5) Transformed data points for one dimensional PCA

→ (6) The reconstruction error for one dimensional PCA.

Solution:-

$$x_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, x_2 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

Step 1: Mean vector: $\bar{x} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$

Step 2: $C = \frac{1}{n} \sum_{i=1}^n \underbrace{(x_i - \bar{x})}_{x_i} \underbrace{(x_i - \bar{x})^T}_{x_i^T}$

18. $(x_1 - \bar{x})(x_1 - \bar{x})^T = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}_{3 \times 1} \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}_{1 \times 3} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

$$(x_2 - \bar{x})(x_2 - \bar{x})^T = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \begin{bmatrix} 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$(x_3 - \bar{x})(x_3 - \bar{x})^T = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$C = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

Step 3: $|C - dI| = 0$ solve this



$d_1 = 3, d_2 = 3, d_3 = 0$

$$(C - 3I)v = 0$$

Step 4: - Unit eigen vectors:

19.

Corresponding to $\lambda_1 = 3$, $w_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ ✓ (or)

Corresponding to $\lambda_2 = 3$, $w_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \checkmark$

Corresponding to $\lambda_3 = 0$, $w_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Note:- If you have different eigenvectors corresponding to same eigenvalues then use any one of the eigen vector to find one dimensional PCA.

Next step:- Projecting data points.

→ Projection of x'_1 on w_1

$$\alpha_1 = x_1'^T w_1 = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}$$

→ projection of x_2^1 on w_1

$$20. \alpha_2 = x_2^{1T} w_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}$$

→ projection of x_3^1 on w_1

$$\alpha_3 = x_3^{1T} w_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = -\sqrt{2}$$

Next step:-

The transformed data points are calculated as:

$$\tilde{x}_i = (x_i^T w_1) w_1 + (x_i^T w_2) w_2 + (x_i^T w_3) w_3$$

→ incorrect

$$\tilde{x}_i = (x_i^T w_1) w_1 + (\bar{x}^T w_2) w_2 + (\bar{x}^T w_3) w_3$$

→ correct

$$\tilde{x}_1 = \underbrace{(x_1^T w_1) w_1}_{\text{these two term vanishes}} + \bar{x}$$

$$= \left(\frac{1}{\sqrt{2}}\right) \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.5 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$$

$$\tilde{x}_1 = \begin{bmatrix} 1.5 \\ 2.5 \\ 2 \end{bmatrix}$$

$$\hat{x}_2 = (x_2^T \omega_1) \cdot \omega_1 + \bar{x}$$

$$\stackrel{21.}{=} \left(\frac{1}{\sqrt{2}}\right) \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 2.5 \\ 2 \end{bmatrix}$$

$$\hat{x}_3 = (x_3^T \omega_1) \cdot \omega_1 + \bar{x}$$

$$= (-\sqrt{2}) \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} +1 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$$

$$\hat{x}_3 = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

Step 4: Reconstruction error:

$$J = \frac{1}{n} \sum \|x_i - \hat{x}_i\|^2$$

$$x_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \hat{x}_1 = \begin{bmatrix} 1.5 \\ 2.5 \\ 2 \end{bmatrix} \Rightarrow x_1 - \hat{x}_1 = \begin{bmatrix} -0.5 \\ -0.5 \\ 1 \end{bmatrix}$$

$$\|x_1 - \hat{x}_1\|^2 = 1.5$$

$$x_2 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \hat{x}_2 = \begin{bmatrix} 1.5 \\ 2.5 \\ 2 \end{bmatrix} \Rightarrow x_2 - \hat{x}_2 = \begin{bmatrix} 0.5 \\ 0.5 \\ -1 \end{bmatrix}$$

$$\|x_2 - \hat{x}_2\|^2 = 1.5$$

$$x_3 = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}, \hat{x}_3 = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \Rightarrow x_3 - \hat{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\|x_3 - \hat{x}_3\|^2 = 0$$

$$\text{Reconstruction error: } J = \frac{1}{3} (1.5 + 1.5 + 0) = 1.$$

$$\boxed{\text{Projected variance} = d_1 = 3}$$

PCA:

(1) zero mean $\rightarrow$

(2) $C = \frac{1}{n} \sum_{i=1}^n x_i x_i^T \rightarrow$ when

(3) eigen value of C

(4) eigen vector of C

(5) projection of x_1 on $w_1 = x_1^T w_1$
 x_2 on $w_1 = x_2^T w_1$
 x_3 on $w_1 = x_3^T w_1$

suppose
 x_1, x_2, x_3

Transformed

(6) $\hat{x}_i = (x_i^T w_1) w_1 + (x_i^T w_2) w_2$

(7) Reconstruction error: $\checkmark$ $\downarrow$ this term will vanish

as we're transforming data point to one dimensional PCA, so we will keep only one Principal Component.

PCA: For our above two problem (given data was not ordered)

24.

(a) we have to find the $\bar{x}$

$$(b) C = \frac{1}{n} \sum_i (\underbrace{x_i - \bar{x}}) (\underbrace{x_i - \bar{x}})^T = \frac{1}{n} \sum_i \underbrace{x_i^T x_i}$$

(c) eigen values of C . $\checkmark$

(d) eigen vectors of C . $\checkmark$

(e) projections of x_1^1 on $w_1 = x_1^{1T} w_1$
 x_2^1 on $w_1 = x_2^{1T} w_1$
 x_3^1 on $w_1 = x_3^{1T} w_1$

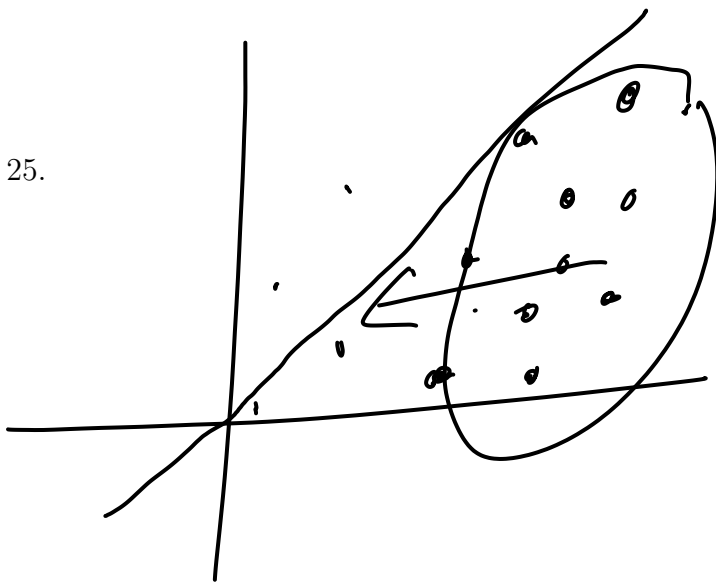
(f) Transfer (for one dimensional PCA)

$$\hat{x}_1 = (x_1^{1T} w_1) w_1 + \bar{x}$$

$$\hat{x}_2 = (x_2^{1T} w_1) w_1 + \bar{x}$$

$$\hat{x}_3 = (x_3^{1T} w_1) w_1 + \bar{x}$$

25.



→ AQ 7-9

new data set: $x_i' = [x_i, 1]$

$$\bar{x}' = [\bar{x}, 1]$$

center this data:

$$x_i' - \bar{x}' = [x_i - \bar{x}, 0]$$

it has no variance.

new covariance matrix:

$$C' = \begin{bmatrix} C & 0 \\ 0 & 0 \end{bmatrix}$$

Rotation is an orthogonal transformation.

26.

$$(R)$$

$$R^T R = I$$

$$(C)$$

new
covariance
matrix

$$C' = R C R^T$$

Let A be a real symm. matrix: $A = A^T$

suppose (v) & (w) are two eigenvectors with
eigen values λ, μ

$$Aw = \mu w$$

$$v^T (Aw) = v^T (\mu w) = \mu v^T w$$

$$\Rightarrow \text{L.H.S. } v^T (Aw) = (v^T A) w = (A^T v)^T w$$

$$= (Av)^T w$$

$$= (\lambda v)^T w$$

$$= \lambda v^T w$$

$$\text{R.H.S.} = \mu v^T w$$

$$\lambda v^T w = \mu v^T w$$

27.

$$\Rightarrow (\lambda - \mu) v^T w = 0$$

if $\lambda \neq \mu$,

$$v^T w = 0$$

↓
eigenvectors are
orthogonal.

28.

29.

30.